

# COVERT STUDENT MENTAL HEALTH SCREENING SYSTEM

## A Research-Backed Indirect Assessment Framework

**Status:** Research/Academic Project | Non-Diagnostic Screening Tool

**Target:** University/College Students (18-25)

**Purpose:** Early detection of mental health patterns requiring support

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## EXECUTIVE SUMMARY

This screening system uses **indirect, behavioral, and cognitive indicators** to identify students at risk for mental health conditions without asking obvious questions about mental health. The design resists social desirability bias through:

- **Scenario-based reasoning** instead of yes/no questions
  - **Embedded cognitive patterns** revealing rumination, avoidance, and hopelessness
  - **Forced choice dilemmas** that reveal values and priorities
  - **Circadian & behavioral anchors** that expose sleep disruption and energy patterns
  - **Narrative inconsistencies** that indicate defensive masking
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## DESIGN PHILOSOPHY

### Core Principles

1. **Psychological Stealth:** Questions feel like lifestyle/preference surveys
2. **Pattern-Based Detection:** Single questions cannot reveal intent; only patterns matter

3. **Resistance to Faking:** Consistent deception requires cognitive load across 32 items
4. **Behavioral Markers:** Sleep, decision-making, social avoidance, rumination patterns reveal truth
5. **Dual Pathways:** Questions can be answered defensively but reveal intent through inconsistencies
6. **No Moral Framing:** All answers are equally “acceptable” and neutral

### Why Indirect Assessment Works

**Research Evidence:** - Social desirability bias causes 20-40% underreporting of depression and anxiety symptoms (Latkin et al., 2017) - Implicit Association Tests bypass conscious self-presentation (Greenwald et al., 1998) - Daily behavioral patterns (sleep, avoidance, rumination) are more reliable than self-report (Place et al., 2017) - Circadian disruption is a biological marker of depression and anxiety (Walker et al., 2020) - Rumination patterns distinguish burnout from depression (Gossmann et al., 2023)

### Three Detection Levels

| Level             | Method                                          | Examples                                            |
|-------------------|-------------------------------------------------|-----------------------------------------------------|
| <b>Explicit</b>   | Direct reporting of specific behaviors/feelings | Sleep schedule, energy patterns, procrastination    |
| <b>Implicit</b>   | Unconscious choices revealing cognition         | Time perception, decision-making, future thinking   |
| <b>Behavioral</b> | Patterns showing coping style                   | Avoidance choices, social withdrawal, perfectionism |

## THE 32 SCREENING QUESTIONS

### SECTION A: DAILY ROUTINES & ENERGY LEVELS

*Detects sleep disruption, circadian misalignment, fatigue, and motivation loss*

**Q1: Weekend Sleep Compensation Question:** “On a typical weekend, compared to weekdays, you tend to sleep...” - A) About the same amount ( $\pm 1$  hour) - B) 2-3 hours longer - C) Much longer (4+ hours) - D) Less than on weekdays

**Question Type:** Forced choice (ordinal)

**Psychological Construct:** Circadian rhythm disruption; chronic sleep debt

**Mental Health Signals:** - High circadian misalignment (options B, C) → Depression, anxiety, burnout - Chronic sleep restriction with compensation → Sleep disorder, mood dysregulation

**Why It Resists Faking:** - Students don't realize this reveals circadian disruption - Lying requires calculating weekend sleep patterns; most won't bother - Even "faking" by saying "A" admits to sleep restriction if true - Circadian disruption is **involuntary**—difficult to fake consistently

**Resistance Logic:** - Faking "A" (same sleep): Requires also lying on energy questions later (inconsistency detection) - Faking "C" (compensation): Requires admitting poor weekday sleep (circular admission) - Cannot lie without creating pattern breaks

**Weight:** HIGH (circadian disruption is biological marker of mood disorders)

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**Q2: Energy During the Day Question:** "Which statement best describes your typical energy throughout the day?" - A) I start strong in the morning and maintain energy through afternoon/evening - B) I'm slow to start, but hit my stride by mid-morning and keep going - C) I have energy bursts at unpredictable times; it's hard to predict my pattern - D) I feel most tired in the afternoon/evening, regardless of sleep

**Question Type:** Forced choice (lifestyle descriptor)

**Psychological Construct:** Circadian phase alignment; fatigue patterns; depression symptom

**Mental Health Signals:** - Option D → Depression (morning-dominant mood symptoms are severe marker) - Option C → Anxiety, irregular coping (unpredictable patterns = dysregulation) - Option A → Baseline/healthy

**Why It Resists Faking:** - Requires student to lie consistently across 2+ weeks of daily data - Most students won't track this precisely to lie; they answer habitually - Detecting option D is especially reliable (depressed students often deny it verbally but admit it when pattern-based)

**Resistance Logic:** - If Q1 shows high weekend compensation + Q2 shows option D → High confidence depression signal - Students cannot fake both consistently

**Weight:** HIGH

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**Q3: When Do You Feel Most Alert? Question:** “You feel most mentally sharp and alert at...” - A) Early morning (before 8 AM) - B) Mid-morning (8 AM - 12 PM) - C) Afternoon (12 PM - 5 PM) - D) Late evening (after 8 PM)

**Question Type:** Forced choice (chronotype descriptor)

**Psychological Construct:** Chronotype; circadian phase; depression phase markers

**Mental Health Signals:** - Option D (delayed chronotype) → Higher risk for depression, anxiety (validated in Chinese college students; Wang et al., 2022) - Combination of D + high weekend compensation (Q1) → Strong depression indicator - Option A + depression-typical answers elsewhere → Severe depression (phase advance)

**Why It Resists Faking:** - Students don’t realize chronotype is linked to mental health - They answer based on actual alertness, not what’s “healthy” - Feels like a neutral personality question

**Resistance Logic:** - Delayed chronotype + irregular sleep + high weekend compensation = high confidence indicator - Cannot be faked without changing actual sleep patterns

**Weight:** MEDIUM-HIGH (chronotype is biological marker)

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**Q4: Your Typical Bedtime Variability Question:** “How consistent is your bedtime throughout the week?” - A) Very consistent—within 1 hour every night - B) Mostly consistent—within 1-2 hours on most nights - C) Quite variable—anywhere from 11 PM to 2 AM - D) Extremely variable—sometimes 2 AM, sometimes 4 AM, no real pattern

**Question Type:** Forced choice (behavioral consistency)

**Psychological Construct:** Sleep hygiene; impulse control; emotional regulation

**Mental Health Signals:** - Options C, D → Anxiety, poor coping, circadian disruption, possible substance use - Option D especially → Unstable mood regulation, possible bipolar risk, severe anxiety

**Why It Resists Faking:** - Students are often unaware of their actual consistency - If they say “A” but later admit to 3 AM bedtimes, we catch inconsistency - Requires living the lie (going to bed at consistent times) to fake authentically

**Resistance Logic:** - Cross-reference with Q1, Q2, Q3 - If Q4 = D and Q1 = A (no weekend compensation), student is likely lying about Q4 - Inconsistency flags defensive masking

**Weight:** MEDIUM-HIGH

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**Q5: Time Perception Speed Question:** “How do you perceive time passing in your day?” - A) Time flows quickly; I’m often surprised how fast the day went - B) Time feels normal—I can roughly estimate how much has passed - C) Time drags occasionally, especially in afternoon/evening - D) Time often feels slow or stuck; hours feel long

**Question Type:** Forced choice (subjective perception)

**Psychological Construct:** Depression symptom (time dilation); anxiety symptom (time pressure); dissociation; anhedonia

**Mental Health Signals:** - Option D → Depression, anxiety, or dissociation - Combination with other depression indicators → High severity - Option A + depression elsewhere → Possible manic/hypomanic episode or avoidance

**Why It Resists Faking:** - This is a highly subjective, non-obvious question - Students don’t realize it reveals depression/dissociation - Lying requires maintaining false time perception in their head (cognitive load)

**Resistance Logic:** - Option D + Q1 (high compensation) + Q2 (option D - afternoon fatigue) → Very high depression probability - Cannot maintain this pattern lie

**Weight:** MEDIUM

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## SECTION B: DECISION-MAKING & PROCRASTINATION

*Detects avoidance, perfectionism, indecision, rumination, and cognitive distortion*

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**Q6: How You Handle Task Deadlines Question:** “When you have an assignment or task due, you typically...” - A) Start immediately or within a day; completion feels manageable - B) Start a few days before; some pressure, but you find your rhythm - C) Feel stuck starting; you often begin the day before - D) Delay until crisis mode; the deadline pressure is what gets you moving

**Question Type:** Forced choice (procrastination pattern)

**Psychological Construct:** Avoidance coping; anxiety about task initiation; executive dysfunction (depression/ADHD); emotion regulation through deadlines

**Mental Health Signals:** - Option D → Anxiety, possibly ADHD, or mood dysregulation (emotional regulation through deadline pressure) - Option C →

Perfectionism, anxiety, possible depression (initiation difficulty) - Pattern of D across multiple questions → Possible depression (avoidance is cardinal feature)

**Why It Resists Faking:** - Students answer based on actual behavior - Lying requires changing actual procrastination patterns to maintain consistency - If they say “A” but later admit task avoidance, we catch it

**Resistance Logic:** - Compare with Q7 (decision-making) and Q8 (perfectionism) - If D on this + high perfectionism pressure elsewhere → Anxiety-driven avoidance (different from depression-driven) - If D + low perfectionism elsewhere → Depression-driven avoidance

**Weight:** HIGH (avoidance is cardinal feature of both anxiety and depression)

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**Q7: Making Decisions Question:** “When faced with a decision (what to wear, what to eat, which assignment to prioritize), you typically...” - A) Decide quickly; you trust your judgment - B) Think it through reasonably, then decide - C) Consider options carefully; decision-making takes time - D) Struggle to decide; you often overthink or default to avoiding the decision

**Question Type:** Forced choice (decision-making style)

**Psychological Construct:** Executive function; rumination; indecision (depression); perfectionism/anxiety (over-analysis); cognitive control

**Mental Health Signals:** - Option D → Depression, anxiety, decision avoidance (possible OCD-related patterns) - Option C + low mood elsewhere → Rumination + depression - Option A + depression elsewhere → Possible avoidance of self-awareness or forced optimism

**Why It Resists Faking:** - Decision-making behavior is automatic and habitual - Students don’t realize this reveals rumination and cognitive control - Lying would require changing actual decision patterns

**Resistance Logic:** - Compare with Q6 (procrastination) - If  $Q6 = D + Q7 = D$  → High avoidance across contexts (anxiety or depression indicator) - If  $Q6 = D + Q7 = C$  → Rumination-driven procrastination (more anxiety than depression)

**Weight:** MEDIUM-HIGH

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**Q8: When Things Go Wrong Question:** “When something doesn’t go as planned or you make a mistake, your first thought is typically...” - A) How can I fix this? What’s the next step? - B) I’m frustrated, but I figure out what happened - C) I replay it in my mind, thinking about what I should have done differently - D) I think about how I’m not good at this, or feel ashamed

**Question Type:** Forced choice (post-failure rumination)

**Psychological Construct:** Post-error rumination; self-blame; perfectionism; shame proneness; depression cognitive style

**Mental Health Signals:** - Option D → Depression (self-blame, shame), possibly anxiety + perfectionism - Option C + option D (combined pattern) → Rumination + depression (strong predictor) - Option A → Healthy problem-solving

**Why It Resists Faking:** - This is an automatic, unconscious thought pattern - Students might not be aware of their typical response - They're more likely to answer truthfully about automatic thoughts

**Resistance Logic:** - Option D repeated across Q8, Q9, Q10 (rumination section) → High rumination burden (strong depression/anxiety indicator) - Cannot fake rumination patterns consistently

**Weight:** MEDIUM-HIGH

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**Q9: Evening Mind Activity Question:** "In the evening, when you're winding down or trying to sleep, your mind typically..." - A) Is calm; you can relax relatively easily - B) Has some activity, but you can focus on sleep - C) Is busy with thoughts about your day or tomorrow - D) Cycles through worries, regrets, or concerns repeatedly

**Question Type:** Forced choice (rumination at night; worry intensity)

**Psychological Construct:** Evening rumination; anxiety (worry); sleep-interfering thoughts; insomnia

**Mental Health Signals:** - Option D → High anxiety, rumination burden, insomnia risk - Option C → Moderate anxiety or moderate rumination - Combination of Q8 (shame rumination) + Q9 (evening rumination) = HIGH rumination load (depression risk)

**Why It Resists Faking:** - Night-time mind activity is habitual and hard to control - Students answer based on actual evening experience - Even if they want to seem calm, their actual sleep is disrupted

**Resistance Logic:** - Option D on Q9 + high weekend sleep compensation (Q1) + afternoon fatigue (Q2) = strong insomnia + mood disorder indicator - Cannot maintain false relaxation while admitting sleep issues

**Weight:** HIGH (evening rumination predicts sleep loss and depression escalation)

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**Q10: Thinking About the Future Question:** “When you imagine life next semester, next year, or after graduation, you tend to feel...” - A) Hopeful or excited about possibilities - B) Neutral; you don’t think about it much - C) Uncertain—some possibilities seem good, others concerning - D) Pessimistic; it’s hard to imagine things working out well

**Question Type:** Forced choice (hopelessness; future orientation; anhedonia)

**Psychological Construct:** Hopelessness; anhedonia; depression symptom; anxiety about future; existential uncertainty

**Mental Health Signals:** - Option D → Depression (hopelessness is cardinal symptom), possible suicidal ideation risk - Option B → Anhedonia (inability to imagine future with pleasure) or avoidance - Combination of D with suicidal/self-harm questions (Q27) → HIGH RISK for depression + ideation

**Why It Resists Faking:** - Hopelessness is involuntary; hard to fake genuine hope - Students answering “A” might be lying, but cross-reference with behavioral questions - If they say “A” but later admit evening rumination and procrastination, we catch inconsistency

**Resistance Logic:** - Option D repeated across multiple depression-indicator questions → Very high depression probability - Combination with Q8 (shame) + Q9 (rumination) + Q10 (hopelessness) = **depression triad** (high severity)

**Weight:** HIGH (hopelessness is predictive of depression severity and suicide risk)

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## SECTION C: SOCIAL PERCEPTION & BELONGING

*Detects loneliness, social anxiety, social withdrawal, and disconnection*

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**Q11: Social Observation Question:** “When you’re in a group setting (class, party, dining hall), you typically feel...” - A) Comfortable; you’re engaged or can easily engage if you want - B) Mostly comfortable, though sometimes you feel a bit on the outside - C) Uncertain if you belong; you observe more than participate - D) Like an outsider; it’s hard to feel like part of the group

**Question Type:** Forced choice (social belonging; social anxiety; withdrawal)

**Psychological Construct:** Social connectedness; social anxiety; loneliness; depression-related withdrawal

**Mental Health Signals:** - Option D → Social anxiety, loneliness, depression (social withdrawal) - Option C → Social anxiety, possible avoidant personality patterns - Combination with Q12, Q13 (social questions) → High loneliness/disconnection (depression risk)



**Why It Resists Faking:** - Social comfort is habitual and hard to fake in actual situations - Even if student says “A,” their actual behavior will be observed - They’re answering about felt experience, which is hard to misrepresent unconsciously

**Resistance Logic:** - If  $Q11 = D + Q12$  (social initiation) =  $D + Q13$  (social support) = weak → HIGH loneliness indicator - Combination with mood questions (Q8, Q9, Q10) → Depression with social withdrawal

**Weight:** MEDIUM-HIGH

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**Q12: Social Initiation Question:** “When it comes to reaching out to friends, attending social events, or initiating plans...” - A) You do it naturally; it doesn’t feel like effort - B) You do it, but you prefer when others initiate sometimes - C) You do it less often; you often feel hesitant - D) You rarely do it; you mostly wait for invitations or withdraw

**Question Type:** Forced choice (social motivation; anhedonia; social anxiety; avoidance)

**Psychological Construct:** Social motivation; anhedonia (loss of pleasure in social interaction); avoidance behavior; depression

**Mental Health Signals:** - Option D → Depression (loss of social motivation), social anxiety, or loneliness - Option C → Possible social anxiety, depression, or burnout - Combination of  $D + Q11$  (outsider feeling) +  $Q13$  (weak support) = **isolation pattern** (high depression risk)

**Why It Resists Faking:** - Social motivation is driven by dopamine/reward; hard to fake consistently - If student says “A” but later admits to rarely texting friends, inconsistency is flagged - Requires changing actual social behavior to maintain the lie

**Resistance Logic:** - Option D on Q12 + Option A on Q11 = INCONSISTENT (lying detected) - Option D on both = Strong depression/anhedonia indicator

**Weight:** MEDIUM-HIGH

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**Q13: Social Support Perception Question:** “If you were going through a difficult time, how confident would you be that friends or family would support you?” - A) Very confident; I have people I trust completely - B) Mostly confident; I have a good support network - C) Somewhat confident, but I’m not sure who would really show up for me - D) Not very confident; I feel like I’d have to handle it alone

**Question Type:** Forced choice (perceived social support; trust; loneliness)

**Psychological Construct:** Perceived social support; attachment security; loneliness; social isolation; hopelessness about support

**Mental Health Signals:** - Option D → High loneliness, depression, possible trauma or insecure attachment - Combination with Q11 + Q12 (both withdrawal/outsider) = **isolation confirmed** (very high depression/anxiety risk) - Option C → Some social anxiety or trust issues; moderate risk

**Why It Resists Faking:** - This reflects actual relationship depth, which is hard to fake - If student says “A” but later admits to isolation, inconsistency is caught - Loneliness is involuntary—hard to claim support you don’t feel

**Resistance Logic:** - Option D on Q13 + options C/D on Q11, Q12 = **isolation pattern confirmed** (multiple independent indicators) - This triad is strong depression/suicidality risk factor

**Weight:** MEDIUM-HIGH

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**Q14: Comparing Yourself to Others Question:** “How often do you find yourself comparing your life, achievements, or appearance to others?” - A) Rarely; you focus on your own path - B) Sometimes; it happens, but doesn’t bother you much - C) Fairly often; you notice how you compare to peers - D) Very often; comparison and self-judgment are constant

**Question Type:** Forced choice (social comparison; rumination; perfectionism; low self-esteem)

**Psychological Construct:** Upward social comparison (depression/anxiety driver); rumination; perfectionism; low self-esteem

**Mental Health Signals:** - Option D → Rumination, perfectionism, social anxiety, depression (constant self-comparison is depressive thought pattern) - Combination with Q8 (shame after mistakes) + Q14 = strong rumination + perfectionism pattern

**Why It Resists Faking:** - This is a habitual cognitive pattern—hard to fake without awareness - If student says “A” but admits perfectionism elsewhere, inconsistency is caught - Social comparison is automatic; requires high cognitive load to fake

**Resistance Logic:** - Option D on Q14 + Option D on Q8 (shame rumination) = **perfectionism + rumination confirmed** (burnout/anxiety risk)

**Weight:** MEDIUM

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## SECTION D: COGNITIVE PATTERNS

*Detects rumination, hopelessness, perceived control, catastrophizing, and identity confusion*

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**Q15: Sense of Control Question:** “In your life right now, how much control do you feel you have over what happens to you?” - A) High control; your choices meaningfully shape outcomes - B) Moderate control; some things are in your control, others aren’t - C) Low control; many things feel outside your influence - D) Very low control; most outcomes feel determined by circumstances beyond you

**Question Type:** Forced choice (locus of control; learned helplessness; depression)

**Psychological Construct:** Internal locus of control; learned helplessness (depression core feature); agency; efficacy

**Mental Health Signals:** - Options C, D → Learned helplessness, depression (loss of agency is core depression symptom) - Combination with Q10 (hopelessness) + Q15 (no control) = **hopelessness + helplessness** (high depression severity)

**Why It Resists Faking:** - Perceived control reflects deep-seated beliefs, not easily faked - Even if student claims “A,” their actual behavior reveals true beliefs - Loss of control is involuntary and shows in procrastination (Q6), avoidance, and rumination

**Resistance Logic:** - Option C/D on Q15 + low grades or avoidance behaviors elsewhere = learned helplessness confirmed - Cannot fake belief in control while showing avoidance across questions

**Weight:** MEDIUM-HIGH

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**Q16: “What-If” Thinking Question:** “How often do you find yourself going through scenarios with ‘what if’ thoughts (like ‘what if I fail,’ ‘what if they judge me,’ etc.)?” - A) Rarely; you don’t tend to rehearse worst-case scenarios - B) Sometimes; it happens occasionally - C) Fairly often; you anticipate potential problems - D) Very often; your mind regularly cycles through potential bad outcomes

**Question Type:** Forced choice (catastrophizing; worry; anxiety; rumination)

**Psychological Construct:** Worry/catastrophizing (anxiety symptom); rumination (depression symptom); cognitive distortion; threat sensitivity

**Mental Health Signals:** - Option D → Anxiety, possible GAD, or depression with anxious features - Combination with Q9 (evening rumination) + Q16 = **anxiety rumination confirmed** (high anxiety probability)

**Why It Resists Faking:** - Catastrophic thinking is habitual and involuntary - Students cannot suppress these thoughts without actual anxiety reduction - This cognitive pattern is hard to fake consistently

**Resistance Logic:** - Option D on Q16 + Option D on Q9 (evening rumination) = worry-rumination loop confirmed (high anxiety) - Cross-reference with Q11, Q12 (social withdrawal) to distinguish anxiety-driven vs depression-driven

**Weight:** MEDIUM-HIGH

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**Q17: Who You Are Question:** “How stable is your sense of identity (who you are, what matters to you, your values)?” - A) Quite stable; you have a clear sense of yourself - B) Mostly stable, with some areas of uncertainty - C) Somewhat unclear; you’re not sure about some core aspects - D) Quite unstable; your sense of identity shifts frequently

**Question Type:** Forced choice (identity stability; self-concept clarity; identity diffusion)

**Psychological Construct:** Identity coherence; ego identity vs role confusion (Erikson); emerging adulthood identity crisis; depression/borderline patterns

**Mental Health Signals:** - Option D → Borderline personality features, severe anxiety, depression with identity loss, or emerging adulthood crisis (normal but needs monitoring) - Combination with Q8 (shame) + Q14 (comparison) + Q17 = **identity instability + perfectionism + rumination** (burnout + depression risk)

**Why It Resists Faking:** - Identity stability is reflected in actual behavior consistency - If student says “A” but answers contradict each other on values (Q14 vs Q8), inconsistency is caught - This is deeply felt and hard to fake sustainably

**Resistance Logic:** - Option D on Q17 + high shame (Q8) + high comparison (Q14) = identity-based perfectionism (burnout risk) - Option D on Q17 + isolation (Q11-Q13) = depression with identity loss

**Weight:** MEDIUM

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**Q18: Changing Your Mind Question:** “How often do you change your opinion about important things (beliefs, preferences, goals) based on new information or circumstances?” - A) Regularly; you’re open to revising your views - B) Occasionally; you’ll change your mind if presented with good reasons - C)

Rarely; once you decide something, you stick with it - D) Very rarely; you have difficulty reconsidering your positions

**Question Type:** Forced choice (cognitive rigidity; openness to change; depression/rumination pattern)

**Psychological Construct:** Cognitive flexibility; growth mindset; fixed mindset (depression feature); rumination (inability to shift thinking); rigidity

**Mental Health Signals:** - Option D → Cognitive rigidity, possible OCD features, depression (rumination stuckness), trauma response, or avoidance of change - Combination with Q8 (shame rumination) + Q18 = **rumination stuckness confirmed** (depression feature)

**Why It Resists Faking:** - Cognitive flexibility is automatic; hard to fake in actual conversations - If student says “A” but admits to black-and-white thinking elsewhere, inconsistency is caught

**Resistance Logic:** - Option D on Q18 + Options D on Q8, Q9, Q16 = rumination stuckness (strong depression/rumination load)

**Weight:** MEDIUM

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## SECTION E: STRESS RESPONSE & COPING STYLE

*Detects active vs passive coping, substance use risk, rumination, and behavioral avoidance*

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**Q19: How You Cope with Stress Question:** “When stressed or overwhelmed, you typically...” - A) Take action; exercise, talk to someone, work on the problem - B) Use a combination of action and relaxation (exercise, time with friends, rest) - C) Try to distract yourself (social media, games, etc.) to take your mind off it - D) Withdraw; you need isolation and tend to ruminate alone

**Question Type:** Forced choice (active vs passive coping; rumination; avoidance; social withdrawal)

**Psychological Construct:** Coping style (active/problem-focused vs avoidant/emotion-focused); rumination; behavioral activation vs deactivation (depression feature)

**Mental Health Signals:** - Option D → Depression (withdrawal + rumination), anxiety (avoidance) - Option C + Option D (pattern across stress questions) → Avoidant coping + rumination (high depression/anxiety risk) - Option A → Healthy coping (baseline)

**Why It Resists Faking:** - Stress coping is automatic and habitual - Under actual stress, students revert to genuine coping styles - Cannot fake active coping if actual behavior is avoidant

**Resistance Logic:** - Option D on Q19 + Option D on Q9 (evening rumination) + Q6 (procrastination) = depression coping profile confirmed - Combination is strong depression indicator

**Weight:** HIGH

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**Q20: Substance Use Mention (Alcohol) Question:** “When you socially drink alcohol, your typical pattern is...” - A) You don’t drink, or drink very rarely - B) Occasional social drinking in a normal amount - C) Regular social drinking, sometimes more than intended - D) Frequent drinking, sometimes alone or to manage stress

**Question Type:** Forced choice (substance use pattern; self-medication risk)

**Psychological Construct:** Substance use frequency; self-medication (depression/anxiety feature); drinking to cope

**Mental Health Signals:** - Options C, D → Alcohol use disorder risk, self-medication of mood/anxiety - Combination with Q19 (avoidant coping) + Q20 = self-medication confirmed (high depression/anxiety risk)

**Why It Resists Faking:** - Substance use pattern is observable by others; hard to fake if social drinking is frequent - Students are more honest about patterns they’ve already admitted to friends

**Resistance Logic:** - Option C/D on Q20 + avoidant coping (Q19, Q6) = self-medication behavior (moderate depression/anxiety + risk factor)

**Weight:** MEDIUM

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**Q21: Physical Exercise Question:** “How frequently do you engage in physical activity (exercise, sports, active recreation)?” - A) Several times a week; it’s part of your routine - B) Once or twice a week - C) A few times a month - D) Rarely or never

**Question Type:** Forced choice (behavioral activation vs deactivation; anhedonia; depression)

**Psychological Construct:** Physical activity (depression protective factor); behavioral activation; anhedonia (loss of motivation for previously enjoyed activities)

**Mental Health Signals:** - Option D → Depression (behavioral deactivation, anhedonia), or low motivation/burnout - Combination with Q12 (social withdrawal) + Q21 = **behavioral deactivation confirmed** (depression symptom)

**Why It Resists Faking:** - Exercise frequency is observable; hard to fake - If student says “A” but admits fatigue/low energy elsewhere, inconsistency is caught

**Resistance Logic:** - Option D on Q21 + afternoon fatigue (Q2) + low motivation elsewhere = depression behavioral deactivation confirmed

**Weight:** MEDIUM

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**Q22: Self-Care Consistency Question:** “How consistent are your habits around self-care (showering, grooming, eating regular meals)?” - A) Very consistent; you maintain these habits daily - B) Mostly consistent; you slip occasionally but quickly get back on track - C) Inconsistent; you sometimes neglect these habits - D) Very inconsistent; these habits often slip, especially when stressed

**Question Type:** Forced choice (behavioral neglect; depression symptom; self-care breakdown)

**Psychological Construct:** Self-care behavior; depression symptom (self-neglect); executive function; motivation

**Mental Health Signals:** - Options C, D → Depression (self-care neglect is cardinal symptom), severe stress, or self-harm risk - Combination with Q2, Q6, Q19, Q21 = depression behavioral symptom cluster confirmed

**Why It Resists Faking:** - Self-care consistency is observable (roommates, family notice) - Hard to fake consistent self-care if actually neglecting it - Shows up in appearance and reported stress

**Resistance Logic:** - Option D on Q22 + afternoon fatigue (Q2) + low exercise (Q21) + avoidant coping (Q19) = **depression symptom cluster** (very high confidence)

**Weight:** HIGH

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## SECTION F: SLEEP & TIME PERCEPTION

*Detects insomnia, sleep fragmentation, circadian disruption, and sleep-related mood dysregulation*

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**Q23: Sleep Quality Question:** “How would you describe your typical sleep quality?” - A) Good; you generally sleep through or wake briefly - B) Mostly good; occasional disruption - C) Mixed; you often wake during the night or have trouble falling asleep - D) Poor; you frequently have difficulty sleeping or wake multiple times

**Question Type:** Forced choice (sleep quality; insomnia severity)

**Psychological Construct:** Insomnia; sleep architecture disruption; depression/anxiety sleep symptom

**Mental Health Signals:** - Options C, D → Insomnia, depression, anxiety - Combination with Q1 (high weekend compensation) + Q4 (variable bedtime) + Q23 = **insomnia + circadian disruption confirmed** (high mood disorder risk)

**Why It Resists Faking:** - Sleep quality is directly experienced; hard to fake - If student claims “A” but admits to waking multiple times (Q24), inconsistency is caught

**Resistance Logic:** - Option C/D on Q23 + high weekend compensation (Q1) + evening rumination (Q9) = insomnia + mood disorder confirmed

**Weight:** HIGH

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**Q24: Sleep Continuity Question:** “During the night, you typically...” - A) Sleep through or wake only once briefly - B) Wake once or twice but fall back asleep quickly - C) Wake multiple times; it takes effort to fall back asleep - D) Frequently wake throughout the night; sleep feels very fragmented

**Question Type:** Forced choice (sleep fragmentation; early morning awakening; depression)

**Psychological Construct:** Sleep fragmentation; early morning awakening (depression hallmark); insomnia; PTSD sleep pattern

**Mental Health Signals:** - Options C, D → Insomnia, depression (early morning awakening is severe depression marker), PTSD, or anxiety - Option D + Option D on Q25 (early morning waking) = **early morning awakening pattern** (severe depression indicator)

**Why It Resists Faking:** - Nighttime waking is involuntary; hard to fake - If student says “A” but admits stress/rumination before sleep (Q9), inconsistency is caught

**Resistance Logic:** - Option D on Q24 + Option D on Q25 (early morning waking) = early morning awakening pattern (strong depression severity indicator)

**Weight:** HIGH



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**Q25: Morning Waking Question:** “How do you typically wake in the morning?” - A) Naturally, at a consistent time; you wake feeling relatively alert - B) Naturally, though sometimes groggier than others - C) You often wake earlier than you’d like, and feel groggy - D) You frequently wake very early (4-5 AM) and can’t fall back asleep, or sleep until very late

**Question Type:** Forced choice (early morning awakening; terminal insomnia; sleep phase; depression)

**Psychological Construct:** Early morning awakening (depression hallmark); sleep phase advance; sleep fragmentation; circadian rhythm

**Mental Health Signals:** - Option D (especially 4-5 AM pattern) → Strong depression indicator (early morning awakening is severe symptom) - Combination with Q24 (fragmented sleep) + Q25 = **early morning awakening confirmed** (high depression severity)

**Why It Resists Faking:** - Early morning waking is involuntary and distressing - Cannot fake this; it either happens or doesn’t - One of most reliable depression biomarkers

**Resistance Logic:** - Option D on Q25 (early 4-5 AM) = depression until proven otherwise - Even if student is high-functioning, this is strong biological signal

**Weight:** HIGH

---

**Q26: Napping Behavior Question:** “How often do you nap during the day?” - A) Rarely or never; you don’t need daytime sleep - B) Occasionally; maybe once a week or less - C) A few times a week; you often feel the need for a nap - D) Frequently; you nap most days to manage tiredness

**Question Type:** Forced choice (daytime sleepiness; sleep debt; depression symptom)

**Psychological Construct:** Daytime sleepiness; sleep debt; depression fatigue symptom; narcolepsy/sleep disorder risk

**Mental Health Signals:** - Options C, D → Sleep debt (from insomnia or circadian disruption), depression fatigue, or sleep disorder - Combination with Q1 (high weekend compensation) + Q23 (poor sleep) + Q26 = **sleep debt confirmed** (high mood disorder risk)

**Why It Resists Faking:** - Daytime sleepiness is involuntary - Napping frequency is habitual and observable

**Resistance Logic:** - Option D on Q26 + afternoon fatigue (Q2) + poor sleep (Q23) = sleep debt from insomnia + mood dysregulation confirmed

**Weight:** MEDIUM-HIGH

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## SECTION G: ACADEMIC SELF-TALK & PERFORMANCE

*Detects perfectionism, impostor syndrome, academic anxiety, and achievement pressure*

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**Q27: Academic Pressure Question:** “How do you feel about your academic performance and expectations?” - A) You’re satisfied with your effort and results; you have realistic expectations - B) You care about your grades, but you’re not overly stressed about them - C) You feel pressure to perform well; there’s meaningful anxiety about grades - D) You feel very high pressure; academic performance feels like it determines your worth

**Question Type:** Forced choice (achievement pressure; perfectionism; internalized pressure)

**Psychological Construct:** Academic self-worth; perfectionism; achievement anxiety; family pressure internalization; impostor syndrome

**Mental Health Signals:** - Option D → High perfectionism, family pressure internalization, possible depression (worth = achievement) - Combination with Q8 (shame after mistakes) + Q14 (constant comparison) + Q27 = **perfectionism-anxiety-depression profile** (moderate-high risk)

**Why It Resists Faking:** - Academic pressure is felt internally; hard to fake - If student says “A” but admits perfectionism elsewhere, inconsistency is caught

**Resistance Logic:** - Option D on Q27 + Option D on Q8 (shame) + Option D on Q14 (constant comparison) = perfectionism-driven burnout confirmed

**Weight:** MEDIUM-HIGH

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**Q28: Failure Response Question:** “If you received a low grade on an assignment or test, you would...” - A) Feel disappointed, but quickly move on and focus on doing better next time - B) Feel upset, but understand it as a learning opportunity - C) Ruminates about it for a few days; it affects your mood - D) Feel deeply ashamed or inadequate; it’s hard to move past it

**Question Type:** Forced choice (post-failure rumination; shame sensitivity; perfectionism; depression)

**Psychological Construct:** Failure rumination; shame proneness; depression response to failure; perfectionism; self-blame

**Mental Health Signals:** - Option D → High shame proneness, depression (negative self-evaluation), perfectionism - Combination with Q8 (shame rumination), Q14 (constant comparison), Q27 (high pressure) = **shame-perfectionism-depression triad confirmed**

**Why It Resists Faking:** - Automatic shame response is involuntary - Faking would require lying about emotional reactivity; hard to do consistently

**Resistance Logic:** - Option D on Q28 + Option D on Q8 + Option D on Q27 = shame-perfectionism-depression clearly confirmed (high severity)

**Weight:** MEDIUM-HIGH

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**Q29: Imposter Syndrome Question:** “When you succeed academically or in other areas, you typically feel...” - A) Pride; you recognize your competence - B) Satisfied; you know you worked for it - C) Some doubt about whether you deserve it; luck or help played a role - D) Like a fraud; you’re convinced others will eventually discover you’re not as capable as they think

**Question Type:** Forced choice (impostor syndrome; competence attribution; depression thinking style)

**Psychological Construct:** Impostor syndrome; attribution style (internal vs external locus); depression cognitive style; perfectionism

**Mental Health Signals:** - Option D → Impostor syndrome, depression (negative self-evaluation despite evidence), perfectionism-driven anxiety - Combination with shame questions (Q8, Q28) + perfectionism (Q27) = **perfectionism-impostor syndrome-depression profile**

**Why It Resists Faking:** - Impostor feelings are ego-syntonic (feel true to the person); hard to fake - If student says “A” but admits doubt elsewhere, inconsistency is caught

**Resistance Logic:** - Option D on Q29 + Option D on Q8, Q28 = impostor syndrome + shame + perfectionism confirmed (burnout/depression risk)

**Weight:** MEDIUM

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## SECTION H: PLEASURE, REWARD, & NUMBNESS

*Detects anhedonia (loss of pleasure), depressive flatness, and emotional numbness*

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**Q30: Activities You Enjoy Question:** “Are there activities or things you typically find enjoyable (hobbies, hanging out, games, food, etc.)?” - A) Yes, several; you look forward to them and they bring genuine pleasure - B) Yes, some; you enjoy them when you do them, but don’t always feel motivated - C) A few things, but you don’t get as much out of them as you used to - D) Not really; even things that used to bring joy feel kind of flat or pointless

**Question Type:** Forced choice (anhedonia; loss of pleasure; depression symptom)

**Psychological Construct:** Anhedonia (cardinal depression symptom); pleasure capacity; reward sensitivity; depression severity

**Mental Health Signals:** - Options C, D → Anhedonia, depression (cardinal symptom) - Option D + Option D on social questions (Q12, Q19) + Option D on Q10 (hopelessness) = **severe depression with anhedonia** (highest risk)

**Why It Resists Faking:** - Anhedonia is involuntary; cannot fake pleasure if not felt - Faking would require maintaining false enjoyment; cognitively difficult

**Resistance Logic:** - Option D on Q30 + Option D on Q10 (hopelessness) + Option D on Q12 (social withdrawal) = **depression with anhedonia confirmed** (high severity indicator)

**Weight:** HIGH

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**Q31: Emotional Range Question:** “Over the past few weeks, how would you describe your emotional range?” - A) You experience a full range of emotions throughout the week - B) You mostly feel okay, with some variation - C) Your emotions feel muted; you don’t feel as much intensity as usual - D) You feel emotionally numb or flat; even things that should evoke emotion don’t

**Question Type:** Forced choice (emotional numbness; flatness; depression; dissociation)

**Psychological Construct:** Emotional blunting (depression symptom); depressive flatness; dissociation; trauma response

**Mental Health Signals:** - Options C, D → Depression with emotional numbing, dissociation, or trauma response - Option D + Option D on Q30 (anhedonia) = **depression with emotional blunting/flatness confirmed** (severe depression)

**Why It Resists Faking:** - Emotional numbness is deeply felt and involuntary - Cannot fake emotional intensity if not experienced

**Resistance Logic:** - Option D on Q31 + Option D on Q30 + Option D on Q10 = **severe depression with anhedonia + emotional flatness** (very high severity)

**Weight:** HIGH

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**Q32: Life Satisfaction Question:** “Overall, how satisfied are you with your life right now?” - A) Quite satisfied; life feels meaningful and worth living - B) Mostly satisfied; there are challenges, but generally good - C) Moderately satisfied; mixed feelings; some good parts, some hard parts - D) Quite dissatisfied; it’s hard to see the value or meaning in your life right now

**Question Type:** Forced choice (life satisfaction; meaning; suicidal ideation screening)

**Psychological Construct:** Overall life satisfaction; meaning-making; suicidal ideation (mild screening); depression severity

**Mental Health Signals:** - Option D → Depression, possible suicidal ideation, hopelessness, meaninglessness (severe risk) - Combination with isolation (Q11-Q13) + hopelessness (Q10) + anhedonia (Q30) + emotional numbness (Q31) + Option D on Q32 = **SEVERE DEPRESSION + SUICIDAL RISK**

**Why It Resists Faking:** - Life satisfaction reflects deep emotional state; hard to fake - If student says “A” but admits anhedonia/numbness elsewhere, inconsistency is caught

**Resistance Logic:** - Option D on Q32 + Option D on Q10, Q30, Q31 = very high depression severity with suicidal ideation risk - **REQUIRES IMMEDIATE REFERRAL**

**Weight:** HIGH (suicidal ideation screening)

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## SCORING FRAMEWORK

### Risk Scoring Algorithm

**Step 1: Domain Scoring** Calculate raw score for each domain:

| Domain                                 | Questions                              | Weight  | Scoring                              |
|----------------------------------------|----------------------------------------|---------|--------------------------------------|
| <b>Sleep &amp; Circadian</b>           | Q1, Q2, Q3, Q4, Q5, Q23, Q24, Q25, Q26 | 9 items | Assign points:<br>A=0, B=1, C=2, D=3 |
| <b>Procrastination &amp; Avoidance</b> | Q6, Q7, Q19, Q21, Q22                  | 5 items | A=0, B=1, C=2, D=3                   |
| <b>Rumination &amp; Cognition</b>      | Q8, Q9, Q10, Q14, Q16, Q18             | 6 items | A=0, B=1, C=2, D=3                   |
| <b>Perfectionism &amp; Shame</b>       | Q27, Q28, Q29                          | 3 items | A=0, B=1, C=2, D=3                   |

| Domain                                    | Questions     | Weight  | Scoring            |
|-------------------------------------------|---------------|---------|--------------------|
| <b>Social Isolation &amp; Loneliness</b>  | Q11, Q12, Q13 | 3 items | A=0, B=1, C=2, D=3 |
| <b>Anhedonia &amp; Emotional Numbness</b> | Q30, Q31, Q32 | 3 items | A=0, B=1, C=2, D=3 |
| <b>Decision &amp; Control</b>             | Q15, Q17, Q20 | 3 items | A=0, B=1, C=2, D=3 |

### Step 2: Calculate Subscale Scores

$\text{Sleep\_Score} = (Q1 + Q2 + Q3 + Q4 + Q5 + Q23 + Q24 + Q25 + Q26) / 9 \times 100$   
 $\text{Avoidance\_Score} = (Q6 + Q7 + Q19 + Q21 + Q22) / 5 \times 100$   
 $\text{Rumination\_Score} = (Q8 + Q9 + Q10 + Q14 + Q16 + Q18) / 6 \times 100$   
 $\text{Perfectionism\_Score} = (Q27 + Q28 + Q29) / 3 \times 100$   
 $\text{Isolation\_Score} = (Q11 + Q12 + Q13) / 3 \times 100$   
 $\text{Anhedonia\_Score} = (Q30 + Q31 + Q32) / 3 \times 100$   
 $\text{Control\_Score} = (Q15 + Q17 + Q20) / 3 \times 100$

### Step 3: Calculate Composite Risk Scores    **Depression Risk Score:**

$\text{DepressionRisk} = (\text{Sleep\_Score} \times 0.25) + (\text{Anhedonia\_Score} \times 0.30) +$   
 $(\text{Rumination\_Score} \times 0.20) + (\text{Isolation\_Score} \times 0.15) +$   
 $(\text{Avoidance\_Score} \times 0.10)$

### **Anxiety Risk Score:**

$\text{AnxietyRisk} = (\text{Sleep\_Score} \times 0.20) + (\text{Rumination\_Score} \times 0.25) +$   
 $(\text{Perfectionism\_Score} \times 0.30) + (\text{Avoidance\_Score} \times 0.15) +$   
 $(\text{Control\_Score} \times 0.10)$

### **Burnout Risk Score:**

$\text{BurnoutRisk} = (\text{Perfectionism\_Score} \times 0.35) + (\text{Rumination\_Score} \times 0.25) +$   
 $(\text{Avoidance\_Score} \times 0.20) + (\text{Sleep\_Score} \times 0.10) +$   
 $(\text{Control\_Score} \times 0.10)$

### **Loneliness/Isolation Risk Score:**

$\text{LonelinessRisk} = (\text{Isolation\_Score} \times 0.50) + (\text{Anhedonia\_Score} \times 0.25) +$   
 $(\text{Avoidance\_Score} \times 0.15) + (\text{Control\_Score} \times 0.10)$

### Step 4: Overall Risk Classification

$\text{Total\_Risk\_Score} = (\text{DepressionRisk} \times 0.35) + (\text{AnxietyRisk} \times 0.30) +$   
 $(\text{BurnoutRisk} \times 0.20) + (\text{LonelinessRisk} \times 0.15)$

Risk\_Level:  
0-20 = Green (Low Risk)  
21-40 = Yellow (Moderate Risk)  
41-60 = Orange (High Risk)  
61-100 = Red (Very High Risk - Immediate Referral)

---

### **Pattern-Based Inconsistency Detection**

#### **Consistency Checks (Flag for Defensive Masking) Check 1: Social vs Behavioral Consistency**

IF (Q11 = A AND Q12 = A) AND (Q19 = D OR Q6 = D) THEN  
INCONSISTENCY FLAGGED = "Claims comfort in social situations but shows avoidance behavior"  
Likelihood\_Masking = HIGH

#### **Check 2: Energy vs Sleep Consistency**

IF (Q2 = A OR Q2 = B) AND (Q23 = C OR Q23 = D) THEN  
INCONSISTENCY FLAGGED = "Claims good energy but reports poor sleep"  
Likelihood\_Deficit\_Underreporting = MEDIUM-HIGH  
TRUE\_RISK = Increase by +15 points

#### **Check 3: Hopelessness vs Engagement Consistency**

IF (Q10 = A) AND (Q30 = D OR Q31 = D OR Q32 = D) THEN  
INCONSISTENCY FLAGGED = "Claims hopeful future but reports emotional numbness/dissatisfaction"  
Likelihood\_Denial = HIGH  
TRUE\_RISK = Increase by +20 points

#### **Check 4: Perfectionism-Shame Triad**

IF (Q27 = D) AND (Q28 = D) AND (Q29 = D) THEN  
Perfectionism\_Shame\_Confirmed = TRUE  
Depression\_Risk += 25  
Burnout\_Risk += 30  
Severity\_Assessment = "High perfectionism + shame-based processing"

#### **Check 5: Isolation Triad**

IF (Q11 = D) AND (Q12 = D) AND (Q13 = D) THEN  
Isolation\_Confirmed = TRUE  
Loneliness\_Risk += 40  
Depression\_Risk += 20  
Suicidal\_Ideation\_Risk\_Level = MODERATE

#### **Check 6: Early Morning Awakening (Severe Depression Marker)**

IF (Q24 = D) AND (Q25 = D with time 4-5 AM pattern) THEN  
Early\_Morning\_Awakening\_Confirmed = TRUE  
Depression\_Risk += 35

ALERT\_LEVEL = "Strong biological depression marker"  
Severity = HIGH

Probable Cause Analysis

Based on risk profile combinations, generate probable primary stressor:

| Risk Pattern                                                 | Probable Cause                                      | Confidence |
|--------------------------------------------------------------|-----------------------------------------------------|------------|
| High Perfectionism + High Shame + Moderate Sleep             | Academic Overload + Family Pressure                 | 85%        |
| High Sleep Disruption + High Rumination + Moderate Isolation | Sleep Debt + Stress Rumination Loop                 | 80%        |
| High Isolation + High Anhedonia + Hopelessness               | Loneliness + Emerging Depression                    | 85%        |
| High Perfectionism + High Anxiety + Moderate Avoidance       | Perfectionism-Driven Anxiety + Academic Pressure    | 80%        |
| High Avoidance + High Procrastination + Moderate Energy Loss | Executive Dysfunction + Possible ADHD or Depression | 75%        |
| High Avoidance + High Substance Use + Moderate Sleep         | Self-Medication Behavior (Coping Style)             | 70%        |
| High Burnout + High Perfectionism + High Rumination          | Burnout Syndrome (Academic/Achievement Stress)      | 80%        |



| Risk Pattern                                                      | Probable Cause                       | Confidence |
|-------------------------------------------------------------------|--------------------------------------|------------|
| High Sleep Disruption + High Rumination + Anxiety Markers         | Anxiety Disorder + Insomnia Loop     | 80%        |
| All domains HIGH (especially Anhedonia + Numbness + Hopelessness) | Major Depression (Clinical Severity) | 90%        |

---

## INTERPRETATION LOGIC

### Green (0-20): Low Risk

**Profile:** Student reports stable mood, good sleep, moderate stress, healthy coping, and strong social support.

**Recommendations:** - General wellness promotion (exercise, sleep hygiene, social connection) - Annual screening follow-up - Information on campus mental health resources (preventive)

---

### Yellow (21-40): Moderate Risk

**Profile:** Student shows some stress indicators (perfectionism, mild sleep disruption, occasional rumination) but maintains functioning and engagement.

**Likely Presentations:** - Normal adjustment stress (academic, social, family transitions) - Mild anxiety about achievement - Early burnout signals

**Recommendations:** - Brief psychoeducational intervention (sleep hygiene, stress management, perfectionism awareness) - Offer campus resources (counseling, stress management workshops) - Rescreen in 4-6 weeks - If risk increases to Orange, escalate

---

### Orange (41-60): High Risk

**Profile:** Student shows multiple mental health warning signs (poor sleep, rumination, social withdrawal, anhedonia, perfectionism-shame pattern).

**Likely Presentations:** - Moderate anxiety disorder - Mild-moderate depression - Burnout - Early identity crisis or loneliness

**Recommendations:** - **Mandatory:** Offer brief counseling consultation (screening + triage) - Connect to campus counseling services - If substance use (Q20 = D) + avoidant coping → Substance use screening - If isolation (Q11-Q13 = D) + hopelessness (Q10 = D) → Suicidal ideation screening - Rescreen in 2-3 weeks - If any RED flag indicators present → Escalate immediately

---

### **Red (61-100): Very High Risk**

**Profile:** Student shows severe depression indicators, social isolation, anhedonia, hopelessness, and possible suicidal ideation.

**Likely Presentations:** - Major Depression Disorder (moderate-severe) - Severe Anxiety Disorder - Suicidal ideation (require screening) - Possible substance use disorder (if Q20 = D) - Severe burnout with hopelessness

**IMMEDIATE ACTIONS:** - **DO NOT** give feedback without clinical follow-up - **Trigger:** Contact campus counseling/mental health immediately - **If Q10 = D + Q32 = D + Isolation = D** → Suicide risk assessment protocol - **If unavailable:** Refer to emergency mental health services, crisis line, or ED - **Campus Safety:** May need to notify (per institutional protocol) if imminent risk - Document and maintain confidentiality

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## **RED FLAG INDICATORS**

### **Suicidal Ideation / Imminent Risk**

**Escalate immediately to crisis protocol if:**

1. **Q10 = D (hopelessness about future) + Q32 = D (life not worth living)** = Moderate ideation risk
2. **Q10 = D + Q30 = D + Q31 = D + Q32 = D** (anhedonia + numbness + hopelessness + dissatisfaction) = **HIGH SUICIDE RISK**
3. **Isolation = D (all three Q11, Q12, Q13) + Hopelessness = D** = Social disconnection + hopelessness (protective factors reduced)
4. **Q20 = D (frequent drinking, drinking alone) + Q19 = D (withdrawal/rumination) + Depression Risk > 70** = Substance use + depression + isolation (very high risk)

**Protocol:** - Brief suicide risk assessment (Columbia Protocol or similar) - Mandatory referral to crisis mental health services - Do not delay

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### Self-Harm / Non-Suicidal Self-Injury

**Screen if:** - Red Risk Score + High Shame (Q28 = D, Q29 = D) + Emotional Numbness (Q31 = D) - Self-harm is not directly asked, but combination signals risk

**Action:** Brief screening question not in main survey; refer if positive

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### Severe Substance Use / Addiction Risk

**Screen if:** - Q20 = D (frequent drinking, drinking alone/to cope) + Q19 = D (withdrawal/avoidant coping) + Sleep\_Score > 70 (using alcohol to sleep)

**Action:** Brief substance use screening; refer to SUD services if indicated

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### Eating Disorder / Body Image Concern

**Note:** Current survey does NOT screen for eating disorders; consider adding question in future iteration if needed

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### Severe Academic Dysfunction

**Flag if:** - Q6 = D (severe procrastination) + Q7 = D (decision paralysis) + Control\_Score > 70 (learned helplessness) - Q27 = D (very high academic pressure) + Perfectionism\_Score > 80 + Q28 = D (shame about failure)

**Action:** Recommend academic coaching, disability services evaluation, or counseling support

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## SAFETY PROTOCOLS

### Before Deployment

**1. Institutional Approval** - ☐ IRB approval (if research) or ethics review - ☐ Consent form (age-appropriate, clear about non-diagnosis) - ☐ Clear data protection plan (encrypted storage, limited access, retention policy) - ☐ Backup mental health resources identified and staffed

**2. Clinical Team Preparation** - ☐ Licensed mental health professional (counselor, psychologist, psychiatrist) on-call for Red Flag cases - ☐ Clear escalation protocol trained with all staff - ☐ Crisis hotline numbers updated in system - ☐ Suicide risk assessment protocol established

**3. Technical Safety** - ☐ Secure login (encrypted passwords) - ☐ No collection of personally identifiable information (use ID numbers) - ☐ Results not visible to faculty/administration (only student + designated counselor) - ☐ Automatic backup to secure server

**4. Student Communication (Pre-Survey)** - ☐ Clear disclaimer (see below)  
- ☐ Informed consent (must check box to proceed) - ☐ Immediate resource card (hotlines, campus counseling, crisis text) - ☐ Post-completion resource list

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## ETHICAL SAFEGUARDS

### Student-Facing Disclaimer

**Title:** "Campus Wellness Reflection Survey"

#### Opening Text:

Thank you for taking this survey. This is a brief, confidential wellness and lifestyle reflection to help us understand how students are doing.

#### IMPORTANT:

- This is NOT a medical diagnosis or psychological assessment
- Your responses are confidential and secure
- Results are NOT shared with faculty, parents, or administration
- This is a screening tool only-it identifies patterns, not disorders
- If you're experiencing a crisis, skip to the end for immediate resources

Your responses help us improve campus mental health support and resources.

#### Post-Survey Disclaimer:

Your results are confidential. Based on your responses, we've created a personalized wellness profile (below). If you're struggling, please reach out.

#### IMPORTANT REMINDERS:

- If you feel unsafe or have thoughts of self-harm, contact:  
[Crisis Hotline] or [Campus Counseling] immediately  
Text HOME to 741741 (Crisis Text Line) anytime
- This survey is NOT a diagnosis. Only a mental health professional can diagnose depression, anxiety, or other conditions.
- Counseling is available at [Campus Center]. First visit is free.

## Confidentiality & Data Protection

1. **Student Data Separation:** Link responses to student ID (no names in database)
2. **Access Control:** Only designated counselors + survey administrator can view results
3. **Secure Storage:** Encrypted database; no cloud storage without institutional approval
4. **Retention Policy:** Delete individual responses after 12 months; aggregate data retained for research
5. **No Faculty/Parent Access:** Results are for student + counselor only
6. **Opt-Out Option:** Student can skip survey entirely, no penalty

## When to Recommend Professional Help

| Risk Level                             | Recommendation                                          | Timeline                       |
|----------------------------------------|---------------------------------------------------------|--------------------------------|
| Yellow (21-40)                         | Consider campus counseling (preventive)                 | Within 2-4 weeks               |
| Orange (41-60)                         | Recommend brief counseling assessment                   | Within 1 week                  |
| Red (61-100)                           | Urgent referral to campus counseling or crisis services | Same day                       |
| Red + Isolation Triad + Hopelessness   | Emergency mental health evaluation                      | Same day / Crisis intervention |
| Red + Q20 = D (frequent substance use) | Dual diagnosis assessment (SUD + mental health)         | Within 1-2 weeks               |

## IMPLEMENTATION GUIDE

### Step 1: Preparation Phase (Weeks 1-2)

**Setup:** 1. Identify secure survey platform (Qualtrics, REDCap, or custom build) 2. Conduct IRB/ethics review 3. Train clinical team (2-3 mental health professionals) 4. Create referral partnerships with counseling center 5. Prepare resource cards & post-survey materials

**Staffing:** - Survey Administrator (1 FTE): Manages survey deployment, tracks Red Flags - Clinical Reviewer (0.5 FTE): Reviews Orange/Red cases, coordinates referrals - Counselor On-Call: Available for crisis escalation

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### Step 2: Launch Phase (Week 3)

**Student Communication:** - Email campaign: "Help us understand student wellness" - Posters: "Confidential wellness survey, 10 minutes" - Incentive (optional): Small gift card raffle for participants

**Deployment:** - Online survey link (accessible from multiple devices) - Paper option (for non-digital access) - QR codes in high-traffic areas

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### Step 3: Ongoing Monitoring (Weeks 4+)

**Daily/Weekly:** - Review Red Flag cases (Q10=D + Q32=D, Isolation Triad, Early Morning Awakening) - Reach out within 24 hours: "We noticed some things in your survey that suggest you might benefit from support..." - Offer warm handoff to counseling

**Monthly:** - Aggregate dashboard: % Yellow, Orange, Red (trend analysis) - Identify high-risk cohorts (e.g., engineering students with burnout, first-year students with isolation) - Adjust resources as needed

---

### Step 4: Follow-Up Protocol (for High-Risk Students)

#### If Orange (41-60):

Email to student:

"Thank you for completing the wellness survey. Your responses suggest you might benefit from talking with a counselor about stress management or mood. We've scheduled a brief consultation for you at [time/location]. If this doesn't work, reply to schedule a different time. You can also call [number] to set up your own appointment."

#### If Red (61-100):

Email + Phone call same day:

"Your responses suggest you're dealing with significant stress. Our counseling center wants to meet with you. [Counselor name] will call you today at [time]. If you're in crisis, please call [hotline] or text HOME to 741741."

---

### **Step 5: Feedback to Student**

#### **Yellow (Low-Moderate Risk):**

Based on your responses, you're managing stress relatively well.

Keep an eye on:

- Sleep schedule (try consistent bedtime)
- Exercise (aim for 3x per week)
- Social connection (reach out to friends regularly)

Campus Resources:

- Counseling: [link] (free, confidential)
- Wellness workshop: Stress management [date]

#### **Orange (High Risk):**

Your responses suggest you're experiencing some significant stress or mood challenges. This is common, and support is available:

Key areas to address:

- Sleep disruption: Consider sleep coaching appointment
- Perfectionism: Counselor can help with stress management
- Social connection: Try reaching out or joining a group

NEXT STEP: We recommend meeting with a counselor. Book here [link] or call [number]. First visit is free. You're not alone.

#### **Red (Very High Risk):**

Your responses indicate you're struggling significantly and may benefit from professional support. Please reach out today.

IMMEDIATE RESOURCES:

- Campus Counseling: [number] (call NOW if available)
- Crisis Hotline: [number]
- Crisis Text Line: Text HOME to 741741
- Hospital ER: [address] if in crisis

You don't have to manage this alone. Help is available.

---

## TECHNICAL IMPLEMENTATION

### Sample JSON Structure (Web App)

```
{
  "survey": {
    "title": "Campus Wellness Reflection",
    "version": "1.0",
    "sections": [
      {
        "name": "Daily Routines",
        "questions": [
          {
            "id": "Q1",
            "text": "On a typical weekend, compared to weekdays, you tend to sleep...",
            "type": "forced_choice",
            "options": [
              {"letter": "A", "text": "About the same amount (±1 hour)", "points": 0},
              {"letter": "B", "text": "2-3 hours longer", "points": 2},
              {"letter": "C", "text": "Much longer (4+ hours)", "points": 3},
              {"letter": "D", "text": "Less than on weekdays", "points": 1}
            ],
            "construct": "Circadian_Rhythm_Disruption",
            "indicators": ["Depression", "Anxiety", "Burnout"],
            "weight": "HIGH"
          }
        ]
      }
    ],
    "scoring": {
      "domains": {
        "Sleep": {
          "questions": ["Q1", "Q2", "Q3", "Q4", "Q5", "Q23", "Q24", "Q25", "Q26"],
          "weight": 0.25
        }
      }
    }
  }
}
```

---

## VALIDATION & ONGOING IMPROVEMENT

### Validation Strategy (Research Phase)

1. **Concurrent Validity:** Correlate with validated scales (PHQ-9, GAD-7, PSQI)



2. **Predictive Validity:** Track which students actually seek counseling; compare with screening predictions
  3. **Test-Retest Reliability:** Readminister to 100 students in 2 weeks; check consistency
  4. **Divergent Validity:** Ensure it does NOT incorrectly flag high-functioning students with normal stress
- 

### Continuous Improvement

**Monthly Review:** - False positive rate: % of Orange/Red students who don't need counseling - False negative rate: % of students in counseling who scored Yellow/Green - Adjust item weights based on predictive accuracy

**Annual Revision:** - Add new items based on emerging student concerns (cost of living, climate anxiety, social media impact) - Remove poorly predictive items - Update scoring thresholds

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## SAMPLE STUDENT PROFILES

### Profile 1: Sarah (Moderate-High Depression Risk)

**Responses:** - Q1 = C (4+ hour sleep compensation) - Q2 = D (afternoon fatigue) - Q6 = D (severe procrastination) - Q8 = D (shame rumination) - Q10 = D (hopeless about future) - Q15 = D (no control) - Q19 = D (withdrawal coping) - Q23 = D (poor sleep quality) - Q30 = D (anhedonia) - Q32 = C (moderately dissatisfied)

**Calculation:** - Depression Risk = 68 → **RED FLAG** - Anhedonia + Rumination + Sleep Disruption + Hopelessness = consistent depression profile - **Primary Concern:** Major depression with possible anxiety

**Action:** Same-day referral to counseling with suicide risk screening

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### Profile 2: James (Burnout Risk)

**Responses:** - Q27 = D (very high academic pressure) - Q28 = D (shame about failure) - Q29 = D (impostor syndrome) - Q8 = D (shame rumination) - Q14 = D (constant comparison) - Q6 = C (procrastination) - Q2 = B (normal energy) - Q30 = B (some enjoyment) - Q32 = B (mostly satisfied)

**Calculation:** - Burnout Risk = 65 → **RED FLAG** - Perfectionism + Shame + Rumination + Academic Pressure = burnout profile - Depression Risk = 35 (moderate; not primary concern)

**Action:** Referral with focus on perfectionism, academic pressure, stress management

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### Profile 3: Aisha (Loneliness + Isolation)

**Responses:** - Q11 = D (outsider feeling) - Q12 = D (no social initiation) - Q13 = D (no support perception) - Q19 = D (withdrawal coping) - Q30 = C (less enjoyment) - Q32 = C (mixed satisfaction) - Q1 = B (2-3 hours compensation; mild) - Q10 = C (uncertain future)

**Calculation:** - Loneliness Risk = 72 → **RED FLAG** - Isolation Triad (Q11-Q13 all D) = confirmed isolation - Depression Risk = 45 (moderate-high)

**Action:** Referral with focus on social connection, loneliness, possible depression screening

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## LIMITATIONS & DISCLAIMERS

### Limitations of This System

1. **Not a Diagnosis:** This is a screening tool. Only mental health professionals can diagnose.
  2. **Self-Report Bias:** Students may intentionally or unintentionally misreport (though design resists this).
  3. **Cultural Differences:** Norms for emotional expression vary by culture; may over-identify or under-identify in some populations.
  4. **Not Individualized Treatment:** Results should trigger referral, not self-treatment.
  5. **Requires Follow-Up:** High-risk students need trained clinician contact, not just automated feedback.
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### Ethical Concerns & Mitigation

| Concern                     | Risk            | Mitigation                                                                  |
|-----------------------------|-----------------|-----------------------------------------------------------------------------|
| Pathologizing normal stress | False positives | Careful threshold calibration; manual review of borderline cases            |
| Privacy breach              | Data exposure   | Encrypted storage; limited access; no parental notification without consent |
| Over-reliance on algorithm  | Missed cases    | Combination of algorithm + manual review by clinician                       |

| Concern            | Risk                   | Mitigation                                                             |
|--------------------|------------------------|------------------------------------------------------------------------|
| Stigma             | Students fear labeling | Framing as “wellness” not “mental illness”; confidential results       |
| Crisis not managed | Student harm           | On-call clinician; crisis hotline resources; clear escalation protocol |

## FINAL RECOMMENDATIONS

### For Research Purposes

1. **Pilot Test** on 50-100 students; collect feedback
2. **Validate** against PHQ-9, GAD-7, PSQI
3. **Refine** items and scoring based on predictive accuracy
4. **Publish** findings (if successful) in peer-reviewed journal
5. **Scale** to larger population with institutional support

### For Campus Implementation

1. **Secure IRB/ethics approval** before launch
2. **Partner with counseling center** for clinical support
3. **Use as early warning system**, not gatekeeping tool
4. **Train student peer supporters** to recognize high-risk indicators
5. **Annual review** of outcomes, false positive/negative rates

### For Future Development

1. Add items on:
  - Financial stress
  - Climate anxiety
  - Social media use
  - Cultural identity & belonging (if diverse campus)
  - Substance use (expanded)
2. Develop **adaptive version** that branches based on responses
3. Create **repeated screening** protocol (semester-based)
4. Integrate with **crisis text line** for real-time intervention

## CONCLUSION

This screening system combines research-backed psychological principles with practical implementation. By using **indirect, scenario-based questions, behavioral markers, and pattern detection**, it identifies students at risk

for mental health challenges while resisting social desirability bias and self-protective responding.

**Key strengths:** - Does not feel like mental health screening - Difficult to fake consistently across 32 items - Captures sleep, rumination, avoidance, shame, and isolation patterns - Generates actionable risk profiles with probable causes - Includes clear safety protocols and ethical safeguards

**Critical success factors:** - **Trained clinical team** to review and act on results - **Clear escalation protocol** for Red Flag cases - **Strong communication** with students (transparency, resources) - **Ongoing validation** and refinement - **Institutional commitment** to follow-up and support

When implemented with care and clinical oversight, this system can serve as an early-warning mechanism for student mental health, allowing campuses to identify and support struggling students before crisis intervention is needed.

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**Document Version:** 1.0

**Last Updated:** December 30, 2025

**Status:** Ready for Pilot Testing

**Questions? Contact:** [Campus Mental Health Director]